

M02 Quick Reference - RC1

Retrieval sheet, not a substitute for practice

DE02-01 Database, table, row, column and query mental model

A database stores related objects such as tables and views. | A query describes the result you want.

DE02-02 SELECT, aliases and LIMIT

Select only columns needed for the question. | Aliases make calculated outputs readable.

DE02-03 WHERE and comparison operators

Use =, <>, >, >=, <, <= carefully. | Text comparisons are quoted.

DE02-04 AND, OR, NOT and parentheses

AND narrows; OR broadens. | Parentheses make precedence explicit.

DE02-05 ORDER BY, DISTINCT and result inspection

ORDER BY changes presentation order, not stored data. | DISTINCT removes duplicate result combinations.

DE02-06 NULL, IS NULL and COALESCE

Use IS NULL / IS NOT NULL. | COALESCE replaces NULL for a specific output purpose.

DE02-07 Arithmetic expressions and safe calculations

Parentheses make calculation order clear. | Use decimal-safe logic for rates.

DE02-08 Aggregate functions

COUNT, SUM, AVG, MIN, MAX are common. | Know whether you are counting rows, IDs or distinct entities.

DE02-09 GROUP BY and grouped grain

Every group is one output bucket. | Grouped result grain should be stated explicitly.

DE02-10 HAVING after aggregation

WHERE filters rows before grouping. | HAVING filters grouped results.

DE02-11 CASE for business categories

Order conditions from specific to general. | Include an ELSE so unmatched rows are visible.

DE02-12 String and date basics

TRIM and case functions can standardize labels. | Date extraction should use date-aware functions where possible.

DE02-13 Simple subqueries and IN/EXISTS awareness

A scalar subquery returns one value. | IN/EXISTS can express membership conditions.

DE02-14 Debugging and validation workflow

Read errors from the first failing clause. | Check row counts before and after filters.